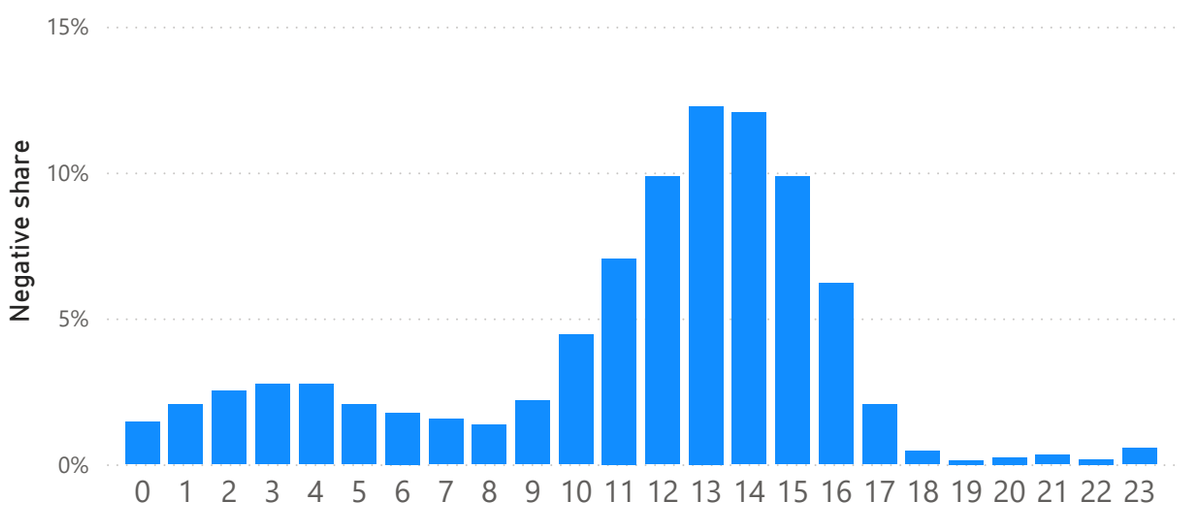
# When is electricity cheap in Germany?

**Hourly day-ahead prices 2018–2026 · Source: Bundesnetzagentur / SMARD.de.**

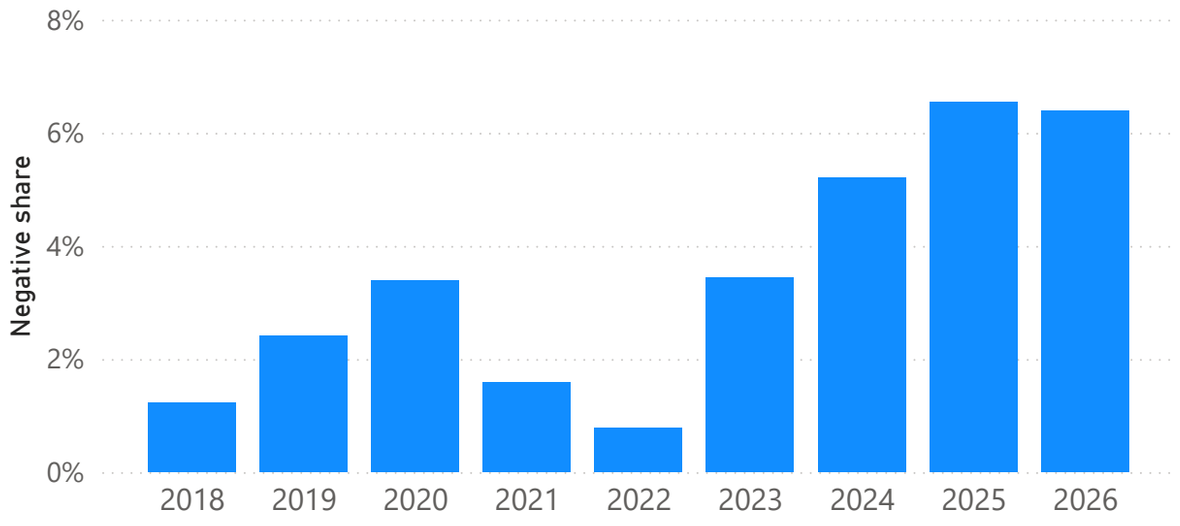
| Negative share | Negative hours | Average price (€/MWh) |
|---|---|---|
| 3.58% | 2,537 | 95.23 |

| | | |
|---|---|---|
| 2018 | 2021 | 2024 |
| 2019 | 2022 | 2025 |
| 2020 | 2023 | 2026 |

## Share of negative-price hours by hour of day (German time)




## Share of hours with negative prices, by year




## Share of negative-price hours by hour and month

| hour | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 2.02% | 3.54% | 3.23% | 0.42% | 0.00% | 0.42% | 0.40% | 0.40% | 1.67% | 1.45% | 0.83% | 3.23% | **1.46%** |
| 1 | 2.42% | 3.54% | 3.23% | 1.25% | 0.40% | 0.42% | 0.40% | 0.81% | 1.67% | 2.54% | 0.83% | 6.85% | **2.03%** |
| 2 | 3.23% | 4.42% | 3.33% | 1.25% | 0.40% | 0.83% | 0.40% | 0.81% | 2.08% | 3.51% | 1.67% | 8.06% | **2.51%** |
| 3 | 4.03% | 4.87% | 3.63% | 1.67% | 0.81% | 0.42% | 0.81% | 0.81% | 1.67% | 3.26% | 2.08% | 8.87% | **2.75%** |
| 4 | 3.63% | 4.87% | 3.23% | 1.25% | 0.81% | 0.42% | 1.21% | 0.81% | 1.67% | 4.35% | 2.08% | 8.47% | **2.75%** |
| 5 | 2.82% | 3.54% | 1.61% | 1.25% | 0.81% | 0.42% | 1.61% | 0.81% | 1.25% | 2.90% | 0.83% | 6.45% | **2.03%** |
| 6 | 2.02% | 2.21% | 2.02% | 1.25% | 0.81% | 0.42% | 1.61% | 0.40% | 0.00% | 3.26% | 0.42% | 6.45% | **1.76%** |
| 7 | 1.61% | 1.77% | 1.21% | 1.25% | 0.81% | 0.42% | 1.61% | 0.81% | 0.00% | 2.90% | 0.42% | 5.65% | **1.56%** |
| 8 | 1.21% | 1.33% | 1.21% | 2.08% | 1.21% | 2.08% | 1.61% | 1.21% | 0.00% | 1.45% | 0.00% | 2.82% | **1.36%** |
| 9 | 1.21% | 1.33% | 2.82% | 3.33% | 2.82% | 5.00% | 2.02% | 2.02% | 0.83% | 2.17% | 0.00% | 2.42% | **2.17%** |
| 10 | 1.21% | 1.77% | 5.24% | 7.50% | 9.68% | 7.08% | 4.84% | 7.26% | 2.50% | 3.26% | 0.00% | 2.82% | **4.44%** |
| 11 | 0.81% | 2.21% | 8.06% | 13.33% | 19.35% | 11.25% | 8.87% | 8.87% | 6.25% | 2.90% | 0.00% | 2.82% | **7.05%** |
| 12 | 1.21% | 2.65% | 11.69% | 20.00% | 22.98% | 18.33% | 13.31% | 12.50% | 9.58% | 3.99% | 0.00% | 2.42% | **9.86%** |
| 13 | 1.61% | 3.10% | 13.31% | 22.50% | 29.03% | 22.92% | 18.15% | 15.73% | 15.00% | 4.71% | 0.00% | 1.61% | **12.27%** |
| 14 | 0.81% | 2.65% | 9.27% | 24.17% | 29.44% | 23.33% | 17.74% | 16.13% | 15.00% | 5.07% | 0.00% | 1.61% | **12.07%** |
| 15 | 1.21% | 1.33% | 5.65% | 20.83% | 26.61% | 19.58% | 15.32% | 14.11% | 10.00% | 2.90% | 0.00% | 1.21% | **9.86%** |
| 16 | 0.81% | 0.88% | 2.42% | 14.17% | 16.94% | 12.92% | 11.29% | 8.06% | 4.58% | 1.45% | 0.00% | 1.21% | **6.20%** |
| 17 | 0.00% | 0.44% | 0.40% | 4.17% | 4.84% | 5.00% | 4.84% | 2.82% | 0.83% | 0.36% | 0.00% | 0.81% | **2.03%** |
| 18 | 0.00% | 0.44% | 0.00% | 1.25% | 0.00% | 2.08% | 0.81% | 0.00% | 0.00% | 0.00% | 0.00% | 0.81% | **0.44%** |
| 19 | 0.40% | 0.44% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.81% | **0.14%** |
| 20 | 0.81% | 0.88% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.42% | 0.81% | **0.24%** |
| 21 | 0.81% | 1.77% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.36% | 0.00% | 0.81% | **0.31%** |
| 22 | 0.40% | 0.44% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 0.36% | 0.00% | 0.81% | **0.17%** |
| 23 | 0.81% | 1.33% | 0.81% | 0.42% | 0.00% | 0.00% | 0.00% | 0.40% | 0.00% | 0.72% | 0.00% | 2.02% | **0.54%** |
| **Total** | **1.46%** | **2.16%** | **3.43%** | **5.97%** | **6.99%** | **5.56%** | **4.45%** | **3.95%** | **3.11%** | **2.25%** | **0.40%** | **3.33%** | **3.58%** |

## What this shows

- Negative prices cluster at 11:00–16:00 in April–June, when solar is strongest.
- In 2025, about 3 in 4 hours at 13:00–14:00 in June were negative.
- In winter, cheap hours move to the night (wind, low demand), e.g. December 00:00–04:00.
- Evenings (17:00–23:00) are almost never negative, in any month.

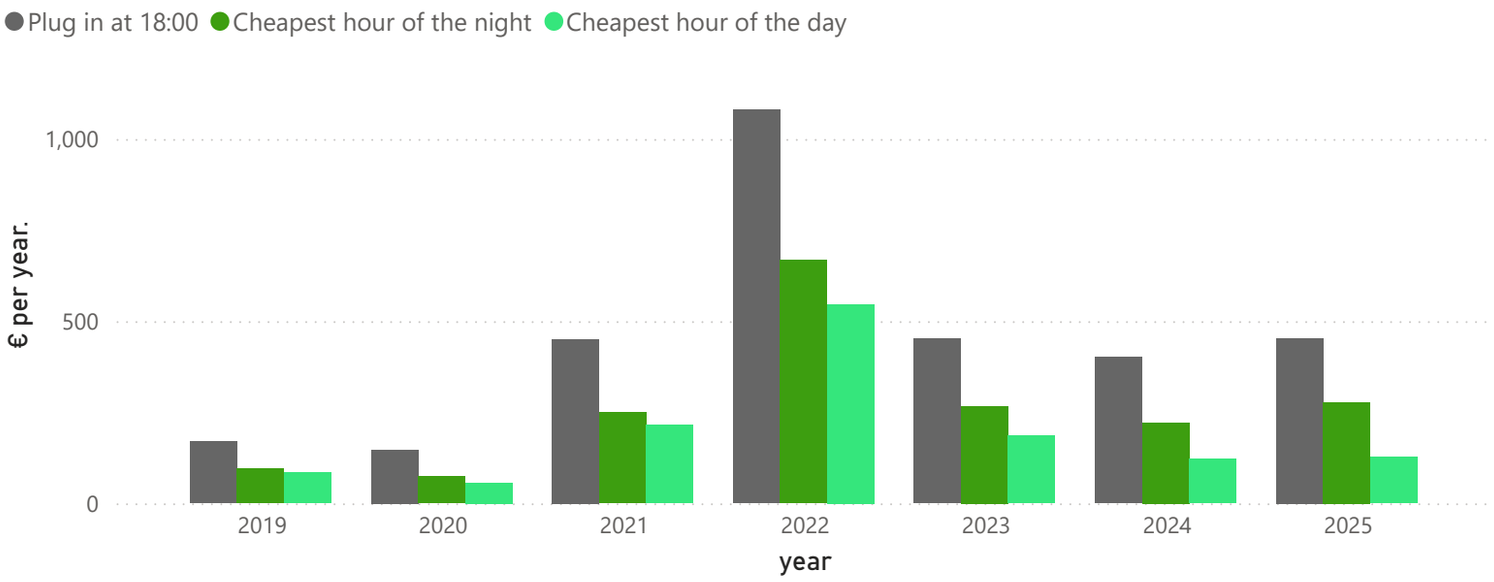
*Select a year above to explore.*

Quick measure

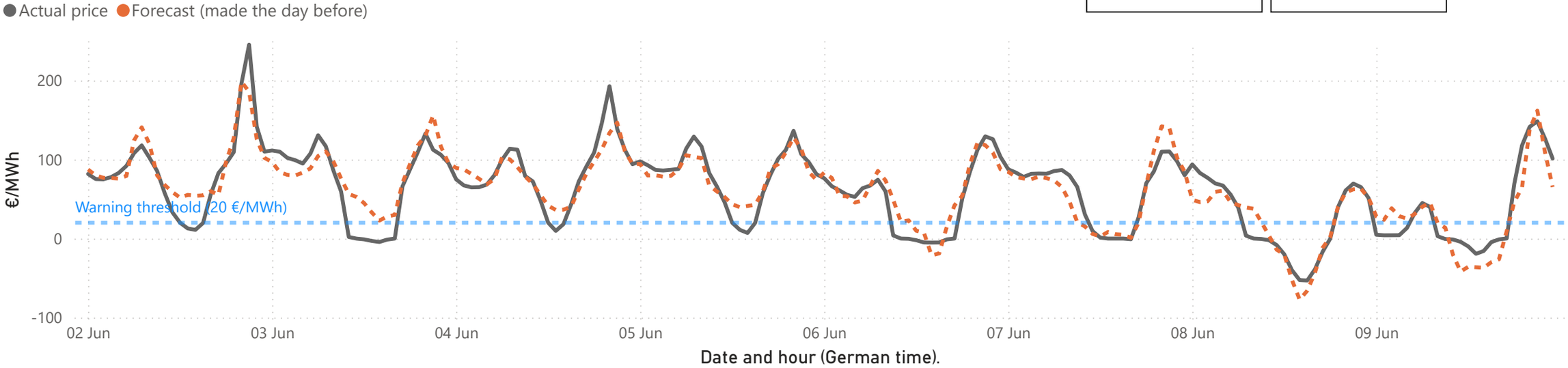
# What is cheap power worth?

EV charging savings and day-ahead price forecast · Sources: SMARD.de, Open-Meteo

Yearly cost of charging an EV, 10 kWh/day (wholesale price, €)




Forecast vs actual price, hourly (€/MWh)




### What this shows

• Charging in the cheapest night hour instead of at 18:00 saved **€176 (39%)** in 2025.

• Charging in the cheapest hour of the day saved **€326 (72%)**.

• The extra saving from daytime charging grew from **€11 (2019)** to **€150 (2025)**: cheap hours are moving to midday.

*Assumptions: 10 kWh/day, car at home 18:00–07:00, wholesale price only.*

Choose a period (e.g. one week)

02-06-2025 — 09-06-2025

Forecast accuracy, test year 2025.

| model | Average error (€/MWh) | Negative hours caught (%) | False alarms (%). |
|---|---|---|---|
| Same as yesterday | 26.0 | 54.3 | 45.7 |
| LightGBM, no weather | 23.8 | 56.0 | 31.4 |
| LightGBM, forecast 1 day ahead | 20.0 | 62.5 | 25.7 |
| LightGBM, actual weather | 19.8 | 62.7 | 26.3 |
| LightGBM, forecast 2 days ahead | 19.6 | 63.4 | 29.1 |

### What this shows

• The forecast is about **25% more accurate** than "same as yesterday" (19.6 vs 26.0 €/MWh).

• Warning below 20 €/MWh, it catches **63%** of negative hours, with **29%** false alarms (vs 54% and 46%).

• Tested with **real 2-day-old weather forecasts**: only information available at 10:00.

Quick measure